Inventory Management and Sales Tracking System with Demand Forecasting for Small Retail Businesses

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In partial fulfillment of the requirements for the degree of

Bachelor of Science in Information Technology

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ABSTRACT

Title : Inventory Management and Sales Tracking System with Demand Forecasting for Small Retail Businesses

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Project Description : This study presents the development of a web-based Inventory Management and Sales Tracking System integrated with demand forecasting to support small retail businesses in managing their operations efficiently. Many small businesses still rely on manual inventory tracking and basic record-keeping, which can result in inaccurate stock levels, overstocking, or stock shortages.

The proposed system allows business owners to monitor inventory levels, record sales transactions, and generate reports in real time. It also incorporates a demand forecasting feature using historical sales data to predict future product demand, helping businesses make informed decisions regarding stock replenishment.

The system was developed using Agile methodology and evaluated based on ISO 25010 software quality standards, including functionality, usability, reliability, performance efficiency, security, maintainability, and portability. Evaluation results indicate that the system improves inventory accuracy, enhances decision

making, and increases overall operational efficiency for small retail businesses

CHAPTER 1: INTRODUCTION

I. Introduction

Effective inventory management is crucial for the success of small retail businesses, as it directly impacts sales, customer satisfaction, and profitability. However, many small retailers continue to use manual methods such as handwritten logs or spreadsheets, which are prone to errors and inefficiencies.

With the advancement of information technology, automated systems can significantly improve inventory control and sales tracking. Additionally, integrating demand forecasting into these systems allows businesses to anticipate customer needs and optimize stock levels.

This study proposes an Inventory Management and Sales Tracking System with demand forecasting to provide a more efficient, accurate, and data-driven solution for small retail businesses.

II. Significance of the Study

This system will benefit the following:

- **Business Owners –** Better control over inventory and sales data
- **Employees –** Simplified transaction recording and stock monitoring
- **Customers** – Improved product availability
- **Small Businesses –** Enhanced decision-making through data insights

III. Statement of the Problem

General Problem

How can a system with demand forecasting improve inventory management and sales tracking for small retail businesses?

Specific Problems

1. How to accurately track inventory levels in real time?
2. How to record and monitor sales transactions efficiently?
3. How to predict future product demand using historical data?
4. How to generate reports to support business decision-making?

IV. Scope and Limitations

Scope

This study focuses on the design and development of a web-based system that integrates inventory management, sales tracking, and demand forecasting for small retail businesses. The system allows users to record product details, monitor stock levels, and track daily sales transactions in real time. It includes features such as adding, updating, and deleting products, generating sales reports, and monitoring low-stock items.

The system also incorporates demand forecasting using historical sales data to predict future product demand. This helps business owners make informed decisions regarding restocking and inventory planning. The forecasting model may use basic techniques such as linear regression or time-series analysis.

The system is intended for use by store owners and authorized staff. It supports user authentication and role-based access. Reports such as daily, weekly, and monthly sales summaries are included, along with visual representations like charts and graphs. The system will be deployed as a web application and will store data in a centralized database.

Limitations

The system is limited to small retail businesses and may not be suitable for large-scale enterprises with complex operations. The accuracy of demand forecasting depends on the availability and quality of historical sales data; limited or inconsistent data may affect prediction results.

The system uses basic forecasting techniques and may not account for external factors such as market trends, seasonal changes, or economic conditions. It does not include advanced features such as multi-branch management, supplier integration, or automated purchasing.

The system requires a stable internet connection for full functionality and may have limited offline capabilities. Security features are limited to basic authentication and may not include advanced cybersecurity measures.

OPERATIONAL DEFINITION OF TERMS

Inventory Management: The process of tracking, organizing, and controlling stock levels of products within a business.

Sales Tracking: The recording and monitoring of sales transactions, including product quantities, prices, and dates.

Demand Forecasting: The process of predicting future product demand based on historical sales data.

Retail Business: A small-scale business that sells goods directly to consumers.

Stock Level: The quantity of a particular product available in inventory.

Low Stock Alert: A system notification indicating that a product has reached a minimum threshold and needs restocking.

Product Management: The process of adding, updating, and removing product information in the system.

Sales Report: A summary of sales data over a specific period (daily, weekly, or monthly).

Forecasting Model: A mathematical or statistical method used to predict future outcomes based on past data.

Linear Regression: A statistical technique used to model and predict relationships between variables, commonly used in forecasting.

Time-Series Analysis: A method of analyzing data points collected over time to identify trends and patterns.

User Authentication: The process of verifying the identity of a user before granting access to the system.

Role-Based Access: A system feature that restricts access based on user roles (e.g., admin, staff).

Database: A structured system used to store, manage, and retrieve data.

Data Visualization: The presentation of data in graphical formats such as charts and graphs.

Transaction: A recorded sales activity involving the purchase of products.

System Accuracy: The degree to which the system provides correct data and reliable forecasts.